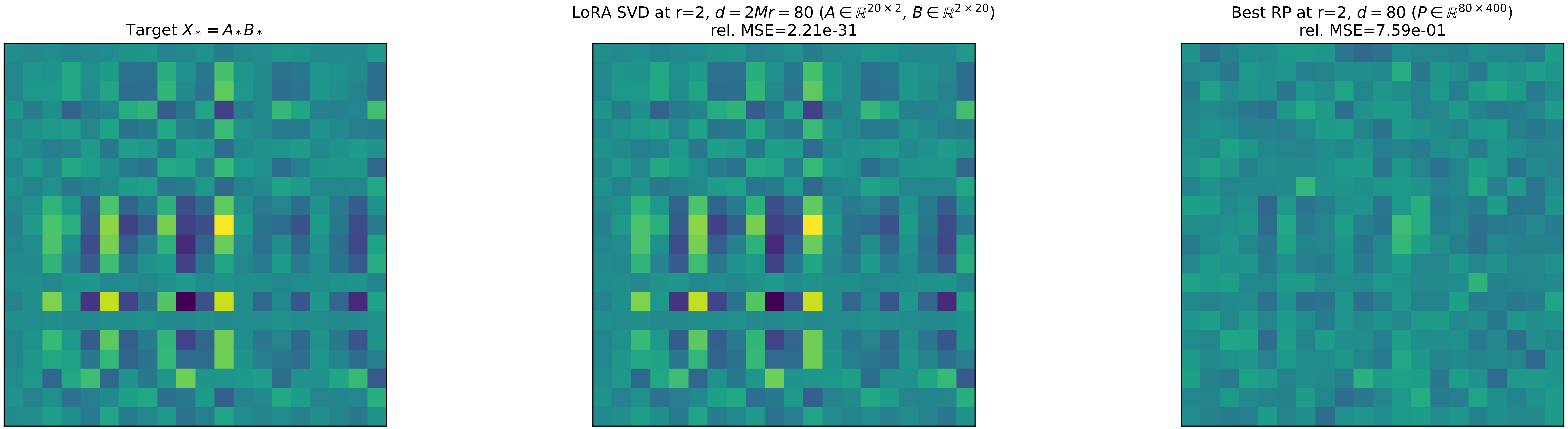
LoRA rank sweep on low-rank shifted benchmark

$M = 20$, $D = M^2 = 400$, target rank=2, benchmark=rastrigin

matched search dims: r=1: d=40, r=2: d=80, r=4: d=160



Setup

Full space: $D = M^2 = 400$.

At each LoRA rank r, all methods search in the same dimension $d = 2Mr$:

LoRA: rank-r factors (A, B), $d = 2Mr$
RP: random projection P, $d = 2Mr$
RB: random blocking groups, $d = 2Mr$

Per-rank search dimensions:
r=1 -> d=40
r=2 -> d=80
r=4 -> d=160

Target rank = 2.

LoRA with r >= target rank can represent the optimum exactly (SVD / factor tuning).

RP/RB use fixed random maps; at each rank they only tune z in the matched d.

At r=2, RB uses $d = 80$ ($D = 400 \to \{0, \ldots, 79\}$).